JavaScript — Day 1 Practice

JavaScript Engine & Execution

> **Goal:** Interview preparation + practical understanding without overwhelming the student.

---

1. Important Interview Questions

JavaScript Engine & Runtime

1. What is a JavaScript Engine?
2. What is a JavaScript Runtime Environment?
3. What is the difference between a JavaScript Engine and Runtime?
4. What happens when JavaScript code is executed?
5. What is the Call Stack?
6. What is the Memory Heap?

Execution Context

7. What is an Execution Context?
8. What is the Global Execution Context?
9. What is a Function Execution Context?
10. What are the two main phases of an Execution Context?
11. What happens during the Creation Phase?
12. What happens during the Execution Phase?
13. What happens when a function is called?
14. What happens when a function finishes execution?

Hoisting

15. What is Hoisting?
16. Are `var`, `let`, and `const` hoisted?
17. What is the difference between `var` hoisting and `let`/`const` hoisting?
18. What is the Temporal Dead Zone (TDZ)?
19. Why does accessing `let` or `const` before declaration cause an error?
20. What happens when `var` is accessed before declaration?

`var`, `let` & `const`

21. What is the difference between `var`, `let`, and `const`?
22. What is the difference between redeclaration and reassignment?
23. Can a `const` variable be reassigned?
24. Can properties of a `const` object be modified?
25. What is block scope?
26. What is function scope?

Strict Mode

27. What is Strict Mode?
28. How do you enable Strict Mode?
29. What happens when you assign a value to an undeclared variable in Strict Mode?

---

2. Predict the Output / Find the Error

For each question:

- Predict the output.
- If there is an error, identify it.
- Explain why.

Q1

```
console.log(a);
var a = 10;
```

Q2

```
console.log(a);
let a = 10;
```

Q3

```
var a = 10;

function test() {
    console.log(a);
    var a = 20;
}

test();
```

Q4

```
var a = 10;

function test() {
    console.log(a);
}
```

```
test();
```

Q5

```
let a = 10;

{
    let a = 20;
    console.log(a);
}

console.log(a);
```

Q6

```
var a = 10;

{
    var a = 20;
}

console.log(a);
```

Q7

```
function test() {
    console.log(a);
    let a = 100;
}

test();
```

Q8

```
function test() {
    console.log(a);
    var a = 100;
}

test();
```

Q9

```
const user = {
    name: "Rahul"
};

user.name = "Amit";

console.log(user.name);
```

Q10

```
const x = 10;
x = 20;

console.log(x);
```

Q11

```
"use strict";

x = 10;

console.log(x);
```

Q12

```
function first() {
    console.log("First");
    second();
}

function second() {
    console.log("Second");
}

first();
```

---

3. Coding Practice

> **Questions only — write the code yourself.**

1. Write a program demonstrating the difference between `var`, `let`, and `const`.

2. Write a program demonstrating hoisting with `var`.
3. Write a program demonstrating the Temporal Dead Zone using `let`.
4. Write a program demonstrating block scope using `let`.
5. Write a program demonstrating the difference between block scope and function scope.
6. Write a program demonstrating that `let` cannot be redeclared in the same scope.
7. Write a program demonstrating that `const` cannot be reassigned.
8. Create a `const` object and demonstrate that one of its properties can still be changed.
9. Write two functions where one function calls another. Observe the order of execution.
10. Write a program with three nested function calls and explain the Call Stack.
11. Write a program containing a global variable and a local variable with the same name. Demonstrate which value is used inside the function.
12. Write a program using `"use strict"` and demonstrate what happens when an undeclared variable is used.

---

4. Mini Interview Scenarios

1. A student says: **"`let` and `const` are not hoisted because they give an error before declaration."** Explain what is actually happening.
2. Why does `var` give `undefined` when accessed before its declaration?
3. Why does `let` give a `ReferenceError` before its declaration?
4. A developer says: **"`const` makes an object completely immutable."** Is this correct? Explain.
5. A function calls another function, which calls a third function. Explain what happens to the Call Stack.
6. Why can a `var` variable declared inside a block be accessible outside the block?
7. Why can a `let` variable declared inside a block not be accessed outside the block?
8. What happens to the Call Stack when a function finishes execution?

---

5. Quick Revision

- What is a JavaScript Engine?
- What is a JavaScript Runtime?
- What is an Execution Context?
- What is the Global Execution Context?
- What is a Function Execution Context?

- What is the Call Stack?
- What is the Memory Heap?
- What is Hoisting?
- What is the Temporal Dead Zone?
- Difference between `var`, `let`, and `const`?
- What are Creation Phase and Execution Phase?
- What is block scope?
- What is function scope?
- What is Strict Mode?
- What causes Stack Overflow?

Day 1 Target

Theory: Understand the concepts, don't memorize definitions.

Output Questions: 12

Coding Practice: 12

Interview Scenarios: 8

Total Practice: 32 questions